Ch.8 Keys

[Step 3: Create data structure]

- Importance of keys
  1. Identify uniquely each row (record)
  2. Enforce integrity
  3. Create table relationship
- Types of keys: candidate; primary; foreign; non-key

Road map thus far for database design:

Step 1: Define mission and objectives (Ch. 5)

Step 2: Analyze current database (Ch. 6)

Step 3: Create data structure (Ch. 7 Tables; 8 Keys; 9 Fields)

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1) Elements of a candidate key

1. Not a multipart field
2. Must contain unique values
3. Cannot contain nulls
4. Cannot breach the organization's security or privacy rules
5. Is not optional
6. Has minimum number of fields to define uniqueness
7. Must uniquely and exclusively identify each record
8. Must exclusively identify each field within a given record (i.e. any other field in a table depends on primary key <u>only</u>, not on other fields as well)
9. Can be modified only in rare or extreme cases

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Find Candidate Keys

Employees

Employee ID	Social Security Number	EmpFirst Name	EmpLast Name	EmpStreet Address	EmpCity	EmpState	EmpZipcode	EmpHome Phone
1000	987-65-9938	Kira	Bently	1204 Bryant Road	Seattle	WA	98157	363-9948
1001	987-65-6531	Katherine	Erlich	101 C Street, Apt. 32	Bellevue	WA	98046	322-6992
1002	987-65-0039	Timothy	Ennis	7402 Kingman Drive	Redmond	WA	98115	527-4992
1003	987-65-1299	Shannon	Black	4141 Lake City Way	Seattle	WA	98136	336-5992
1004	987-65-6529	Susan	Black	2100 Mineola Avenue	Seattle	WA	98115	572-9948
1005	987-65-5583	Estela	Rosales	101 C Street, Apt. 32	Bellevue	WA	98046	322-6992
1006	987-65-1734	Timothy	Sherman	66 NE 120th	Bothell	WA	98216	522-3232

Figure 8.1 *Are there any candidate keys in this table?*

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Find Candidate Keys

Employees

Employee ID	Social Security Number	EmpFirst Name	EmpLast Name	EmpStreet Address	EmpCity	EmpState	EmpZipcode	EmpHome Phone
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1002	987-65-0039	Timothy	Ennis	7402 Kingman Drive	Redmond	WA	98115	527-4992
1003	987-65-1299	Shannon	Black	4141 Lake City Way	Seattle	WA	98136	336-5992
1004	987-65-6529	Susan	Black	2100 Mineola Avenue	Seattle	WA	98115	572-9948
1005	987-65-5583	Estela	Rosales	101 C Street, Apt. 32	Bellevue	WA	98046	322-6992
1006	987-65-1734	Timothy	Sherman	66 NE 120th	Bothell	WA	98216	522-3232

Figure 8.1 *Are there any candidate keys in this table?*

Table Structures

Employees

Employee ID	*CK*
Social Security Number	
EmpFirst Name	*CCK*
EmpLast Name	*CCK*
EmpStreet Address	
EmpCity	
EmpState	
EmpZipcode	
EmpHome Phone	

Figure 8.2 *Marking candidate keys in the EMPLOYEES table structure*

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Are there candidate keys here?

Parts

Part Name	Model Number	Manufacturer Name	Retail Price
Shimka XT Cranks	XT-113	Shimka Incorporated	199.95
Faust Brake Levers	BL / 45	Faust USA	53.79
MiniMite Pump		MiniMite	35.00
Hobo Fanny Pack		Hobo Bike Company	59.00
Diablo Bike Pedals	Mtn-A26	Diablo Sports	129.50
Shimka Truing Stand	SP-100		37.95
Faust Brake Levers	BL / 60	Faust USA	79.95

Figure 8.3 *Can you identify any candidate keys in the PARTS table?*

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No candidate keys? ➔ Create artificial candidate keys

Parts

Part Number	Part Name	Model Number	Manufacturer Name	Retail Price
41000	Shimka XT Cranks	XT-113	Shimka Incorporated	199.95
41001	Faust Brake Levers	BL / 45	Faust USA	53.79
41002	MiniMite Pump		MiniMite	35.00
41003	Hobo Fanny Pack		Hobo Bike Company	59.00
41004	Diablo Bike Pedals	Mtn-A26	Diablo Sports	129.50
41005	Shimka Truing Stand	SP-100		37.95
41006	Faust Brake Levers	BL / 60	Faust USA	79.95

Figure 8.4 *The PARTS table with the artificial candidate key* PART NUMBER

Note: even if we have candidate keys (like, FirstName + LastName), we might create artificial candidate keys (like StudentID) because it is shorter and helps with relationships between tables

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2) Select primary key from candidate keys

Table Structures

Employees

Employee ID	*CK*
Social Security Number	
EmpFirst Name	*CCK*
EmpLast Name	*CCK*
EmpStreet Address	
EmpCity	
EmpState	
EmpZipcode	
EmpHome Phone	

Figure 8.5 *Which candidate key should become the primary key of the EMPLOYEES table? [CCK = composite candidate key]*

1) Has the fewest number of fields
2) Contains parts of table name

- Exclusively identifies the table throughout the database structure and helps establish relationships with other tables
- Uniquely identifies a given record within a table and exclusively represents that record throughout the entire database

1. Each table must have only one primary key
2. Each primary key is unique in the entire database structure except:
   - 1-to-1 relationship
   - Table B is subset of table A

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3) Test a Primary Key

Sales Invoices

Invoice Number	Invoice Date	CustFirst Name	CustLast Name	EmpFirst Name	EmpLast Name	EmpHome Phone
13000	06/15/02	Frank	DeSoto	Estela	Rosales	363-9948
13001	06/15/02	Gregory	Mattson	Katherine	Erlich	322-6992
13002	06/15/02	Carmen	Aguilar	Kira	Bently	527-4992
13003	06/16/02	David	Cunningham	Kira	Bently	336-5992
13004	06/16/02	Carmen	Aguilar	Shannon	Black	572-9948
13005	06/17/02	Frank	DeSoto	Estela	Rosales	322-6992

Figure 8.6 *Does the primary key exclusively identify the value of each field in this table?*

Question: Is **Invoice Number** uniquely Identifying **Invoice Date**? **CustFirst Name**? etc…

Answer: Yes, yes, yes, yes, yes, no.

Ex.: **Invoice Number** indirectly identifies **EmpHome Phone** via **EmpLast Name** ➔ remove field EmpHome Phone from this table.

[There is only one Estela Rosales, who changed their phone, but we updated it in only one place ➔ update inconstancy (violation of the 2nd NF)]

EmpHome Phone should entirely depend on PK, NOT on other fields; here EmpHome Phone depends on EmpFirst Name and EmpLast Name

Table Structures

Sales Invoices

Invoice Number	*PK*
Invoice Date	
CustFirst Name	
CustLast Name	
EmpFirst Name	
EmpLast Name	
Ship Date	
Shipper Name	

Figure 8.7 *The revised SALES INVOICES table with its new primary key*

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Alternate key (AK) and non-key

Table Structures

Employees

Employee ID	*PK*
Social Security Number	
EmpFirst Name	*CAK*
EmpLast Name	*CAK*
EmpStreet Address	
EmpCity	
EmpState	
EmpZipcode	
EmpHome Phone	

- Alternate keys = the rest of candidate keys after choosing a primary key
- Non-key = any other field that is not a primary key or (composite) alternate key

Figure 8.8 *The EMPLOYEES table with designated primary and alternate keys*

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MySQL to designate keys → via a constraint

- CREATE TABLE person
  (**person_id SMALLINT UNSIGNED,**
  fname VARCHAR(20),
  lname VARCHAR(20),
  eye_color ENUM('BR','BL','GR'),
  birth_date DATE,
  street VARCHAR(30),
  city VARCHAR(20),
  state VARCHAR(20),
  country VARCHAR(20),
  postal_code VARCHAR(20),
  CONSTRAINT pk_person PRIMARY KEY (person_id)
  );

- CREATE TABLE favorite_food
  (**person_id SMALLINT UNSIGNED,**
  food VARCHAR(20),
  CONSTRAINT pk_favorite_food PRIMARY KEY (person_id, food),
  CONSTRAINT fk_fav_food_person_id FOREIGN KEY (person_id) REFERENCES person (person_id)
  );